Ear is the organ of hearing. The nerve supplying the ear is the 8th cranial nerve which is also called the auditory nerve or the vestibule cochlear nerve. The ear is divided into 3 parts:

- External ear
- Middle ear
- Internal ear.

External Ear

- Pinna or auricle – collects the sound waves
- External auditory meatus – conveys the vibrations of sound.

Middle Ear

- Ear drum – communicates to the mastoid process
- Eustachian tube - maintaining the pressure of air
- Auditory ossicles – 3 small bones (malleus, incus, stapes).

Mastoid process is the part of the temporal bone lying behind the ear. It is an air space which communicates in the middle ear.

Internal Ear

Consists of cavities called the bony labyrinth and membranous labyrinth. Bony labyrinth consists of 3 parts:

- The vestibule
- Semi circular canals
- Cochlea.

PHYSIOLOGY OF EAR

Sound waves pass along the external auditory canal cause the tympanic membrane to vibrate. The vibrations transmitted through malleus, incus and stapes. By movement of these bones, the vibrations are magnified and then communicated to the vestibular fenestra to the perilymph and to the endolymph in the canal of the cochlea. This stimuli reaches the nerve endings in the organ of corti and conveyed to the brain by auditory nerve.

SKIN

- Largest sense organ is SKIN. The surface area of skin is about 19.4 sq/ft.
- Lymph vessels are absent in epidermis, hair, nail, cartilage, cornea and central nervous system.

The skin covers the body. It consists of dermis and epidermis. It completely covers the body and protect the under lying structure from injury and infection by the bacteria.

EPIDERMIS

This is the outermost thin portion of the skin. No blood vessels are found in this layer. It derives its nutrition from lymph. Nerves are found in this layer. The epidermis consists of four layer of cells. They are.

- The stratum corneum
- The stratum lucidum
- Stratum granulosum and
- the stratum malphigi.

DERMIS

This is situated below the epidermis. It is the most thickest dermis formed by connective tissue which is richly supplied with blood vessels and nerves.

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